

Julia Fancher

The Ohio State University
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EDUCATION

The Ohio State University, Columbus, OH

Ph.D. in Astronomy

expected 2032

M.S. in Astronomy

expected 2028

Syracuse University, Syracuse, NY

May 2026

Bachelor of Science in Physics and Applied Mathematics, Minor in Logic

4.0 GPA

- Renee Crown University Honors Program

AWARDS AND HONORS

NSF Graduate Research Fellow

2026

The Ohio State University Graduate Fellowship

2026

Syracuse University Honors:

- Syracuse University Scholar 2026
- Syracuse University Arts and Sciences Scholar 2026
- Peterfrund Prize 2026
- Astronaut Scholar 2024 & 2025
- Goldwater Scholar 2024
- Euclid Prize 2025
- Emerging Research Fellow, Syracuse University, undergraduate research grant 2023-2026
- Syracuse Leadership Scholar, tuition grant 2022-2026
- SU Physics Coaching Award 2023

RESEARCH EXPERIENCE

Research on magnetic winds conducted under Dr. Todd Thompson

- Determined the effect of changing the magnetic field configuration on stellar spindown rates

Research on tidal disruption events conducted under Dr. Eric Coughlin

- Ran PHANTOM simulations with and without self gravity to isolate its effects on tidal disruption event debris streams; results published as first author
- Characterized the predictions of the CN22 model and their implications on the interpretation of observed data; results published as second author
- Ran PHANTOM simulations to test the predictions of the CN22 model and create a library of realistic stars; results published as first author
- Used TDE simulations with realistic stellar structure to predict the percentage of jetted TDEs; thesis for Renee Crown Honors Program

PUBLICATIONS

- *Predicting the Properties of the Fallback Rate from Tidal Disruption Events: Investigating the Maximum Gravity Model.* **Fancher, J.**, et al. 2025. *Astrophysical Journal*, 990, 2
- *The Peak of the Fallback Rate from Tidal Disruption Events: Dependence on Stellar Type.* Bandopadhyay, A., **Fancher, J.**, et al. 2024. *Astrophysical Journal Letters*, 962, L2
- *On the relative importance of shocks and self-gravity in modifying tidal disruption event debris streams.* **Fancher, J.**, et al. 2023. *Monthly Notices of the Royal Astronomical Society*, 523, 2323

PRESENTATIONS

- *Investigating the Maximum Gravity Model.* **Fancher, J.** Presented at the ESAC workshop in Madrid 2025.
- *The relative importance of shocks and self-gravity in modifying tidal disruption event debris streams.* **Fancher, J.**, et al. Presented at the 243rd AAS conference 2024; undergraduate Chambliss finalist.
- *Tidal Disruption Events.* **Fancher, J.** Presented at the PHANTOM workshop in Melbourne 2024.
- *The relative importance of shocks and self-gravity in modifying tidal disruption event debris streams.* **Fancher, J.** Presented at the Syracuse University Undergraduate Research Fair, fall 2023.
- *A Controlled Set of Numerical Experiments of Stars Destroyed by Supermassive Black Holes.* **Fancher, J.** Presented at the Syracuse University Undergraduate Research Fair, spring 2023.

TEACHING AND MENTORING

<i>Physics Peer Coach</i> , Syracuse University	2023-2026
• Support students in PHY 102, PHY 222, PHY 211, and PHY 365	
<i>Holden Observatory Tour Guide</i> , Syracuse University	2023-2026
• Lead tours of Holden Observatory, including operating the telescope for evening tours	
<i>Society of Physics Students Executive Board Member</i>	2023-2025
• Served as treasurer and secretary during the 2023-2024 and 2024-2025 academic years	
<i>Friends of Inkululeko volunteer</i>	2023-2026
• Headed “Options After Matriculation” project while in Makhanda, South Africa • Assisted in entrepreneurship skills workshop in Makhanda, South Africa • Raise awareness and involvement in Inkululeko’s mission	
<i>Pathway Prep tutor</i> , Syracuse University	2023-2024
• Tutor students in rural Mexico in SAT math	